

UMTC Automation- Fab Integration AMHS SECS Common Specification

Document ID	UMTC Automation- Fab Integration AMHS SECS Common Specification	Created Date	2022/07/07
File Name	Unimicro Automation Fab Integration-AMHS SECS Common Specification Ver 1.0.4.doc	Last Revised	2025/04/08
Version No	Ver 1.0.4	Number of Pages	77

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Revision History

Version	Date	Page	Description	Revised by
1.0.0	2022/07/07	All	1. First Version.	Alex
1.0.1	2022/07/11	19	1. Modify Carrier State Model Add IB SEM &Stock SEM	Alex
1.0.2	2022/07/13	49	1. Delete S2F25,Add CEID160	Alex
1.0.3	2024/04/22	47~49 74 71 67 38~41 50 63	1. Modify Scenario 5.11~5.13 2. Modify HCACK 3. Add RCMD List" NOTRANSFERNOTIFY" 4. Add ProcessState 、CARRIERTYPE 5. 04.02.34 S2F49 Enhanced Remote Command Extension add Carrier Type 6. 05.14 Transport with ID mismatch or ID Read Failed 7. Result Code add 10 : Error But Carrier Already in Dest	Shryu
1.0.4	2025/03/26	64 60 51	1. Result Code add 6 :Source Empty 2. 06.01 CEIDList Add 705 AGV Mode Change Report/ 706 MGV Mode Change Report 3. 05.15 Add AGV/MGV Mode Change Flow	Ben

1. Scope

This specification is used for MCS project as equipment automation common specification. It applies only to the equipment that uses the **SEMI Equipment Communication Standards (SECS)** as the communication protocol. Equipment is passive and MCS is active.

該規範用於項目作為設備自動化通用規範。它僅適用於使用 SEMI 設備通信標準 (SECS) 作為通信協議的設備。設備為被動端, MCS 為主動端。

適用範圍

1. 適用對象 Suitable Object

AMHS 自動搬運及自動倉儲設備

2. 適用區域 Suitable Area

全部 AMHS 自動化搬送

Apply to all AMHS equipment

3. 適用時機 Suitable Time

適用於欣興電子之 AMHS 自動化搬送機台之採購 (包含客戶委託之機台)。

Apply to UNIMICRON AMHS equipment purchasing (include equipment supplied by customer)

通訊規格 Communication Specification

參考文件

SEMI E4 : SEMI EQUIPMENT COMMUNICATIONS STANDARD MESSAGE TRANSFER (SECS-I)

SEMI E5 : SEMI EQUIPMENT COMMUNICATIONS STANDARD II MESSAGE CONTENT (SECS-II)

SEMI E30 : GENERIC MODEL FOR COMMUNICATIONS AND CONTROL OF SEMI EQUIPMENT (GEM)

SEMI E37 : HIGH-SPEED SECS MESSAGE SERVICES (HSMS) GENERIC SERVICES

SEMI E37.1 : HIGH-SPEED SECS MESSAGE SERVICES SINGLE-SESSION MODE (HSMS-SS)

SEMI E88 : Specification for AMHS Storage SEM (Stocker SEM)

SEMI E82 : SPECIFICATION FOR AMHS INTERBAY/INTRABAY AMHS SEM (IBSEM)

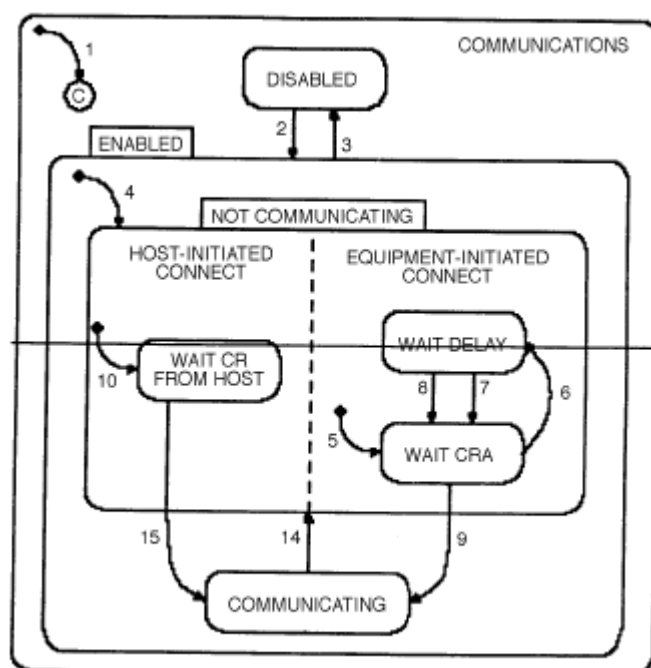
2. 狀態定義(Status Definition)

02.01 Common States

02.01.01 Communication States

There are two major states of SECS communication, DISABLED and ENABLED. The system default state must be user configurable at the equipment (e.g., via a jumper setting or non-volatile memory variable).

Once system initialization has been achieved, the operator shall be able to change the communication state selection at any time via equipment terminal functions or momentary switch. A two-position type switch must not be used due to possible conflict with the system default.



State	Description
DISABLED	In this state SECS-II communication with a host computer is non-existent. If the operator switches from ENABLED to DISABLED, all SECS-II communications must cease immediately. Any messages queued to send shall be discarded, and all further action on any open transactions and conversations shall be terminated. Handling of messages currently being

	transmitted is an issue for lower level message transfer protocols and is not addressed in this standard. The DISABLED state is a possible system default.
ENABLED	ENABLED has two sub states, COMMUNICATING and NOT COMMUNICATING. Whenever communications are enabled, either during system initialization or through operator selection, the substate of NOT COMMUNICATING is active until communications are formally established. Lower-level protocols (such as SECS-I) are assumed to be functioning normally in that they are capable of supporting the communication of SECSII syntax. The ENABLED state is a possible system default.
NOT COMMUNICATING	No messages other than S1,F13, S1,F14, and S9,Fx shall be sent while this sub state is active. The equipment shall discard any messages received from the host other than S1,F13 or S1,F14 (Establish Communications Acknowledge). It shall also periodically attempt to establish communication with a host computer by issuing an S1,F13 until communications are successfully established. However, only one equipment-initiated S1,F13 transaction may be open at any time. The NOT COMMUNICATING state has two AND substates, HOST-INITIATED CONNECT and EQUIPMENT-INITIATED CONNECT, both of which are active whenever the equipment is NOT COMMUNICATING. These two substates clarify the behavior of the equipment in the event that both the equipment and the host attempt to establish communications during the same period of time.
EQUIPMENT-INITIATED CONNECT	This state has two sub states, WAIT CRA and WAIT DELAY. Upon any entry to the NOT COMMUNICATING state, whenever EQUIPMENT-INITIATED CONNECT first becomes active, a transition to WAIT CRA occurs, the Comm Delay timer is set to "expired," and an immediate attempt to send S1,F13 is made.
WAIT CRA	An Establish Communications Request has been sent. The equipment waits for the host to acknowledge the request
WAIT DELAY	A connection transaction failure has occurred. The Comm Delay timer has been initialized. The equipment waits for the timer to expire.
HOST-INITIATED CONNECT	This state describes the behavior of the equipment in response to a host-initiated S1,F13 while NOT COMMUNICATING is active
WAIT CR FROM HOST	The equipment waits for an S1,F13 from the host. If an S1,F13 is received, the equipment attempts to send an S1,F14 with COMMACK = 0.
COMMUNICATING	Communications have been established. The equipment may receive any message from the host, including S1,F13. When the equipment is

	COMMUNICATING, SECS communications with a host computer must be maintained. This state remains active until communications are disabled or a communication failure occurs. If the equipment receives S1,F13 from the host while in the COMMUNICATING sub state, it should respond with S1,F14 with COMMACK set to zero. If the equipment receives S1,F14 from a previously sent S1,F13, no action is required. In the event of communication failure, the equipment shall return to the NOT COMMUNICATING substate and attempt to reestablish communications with the host.
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02.01.02 Control States Model

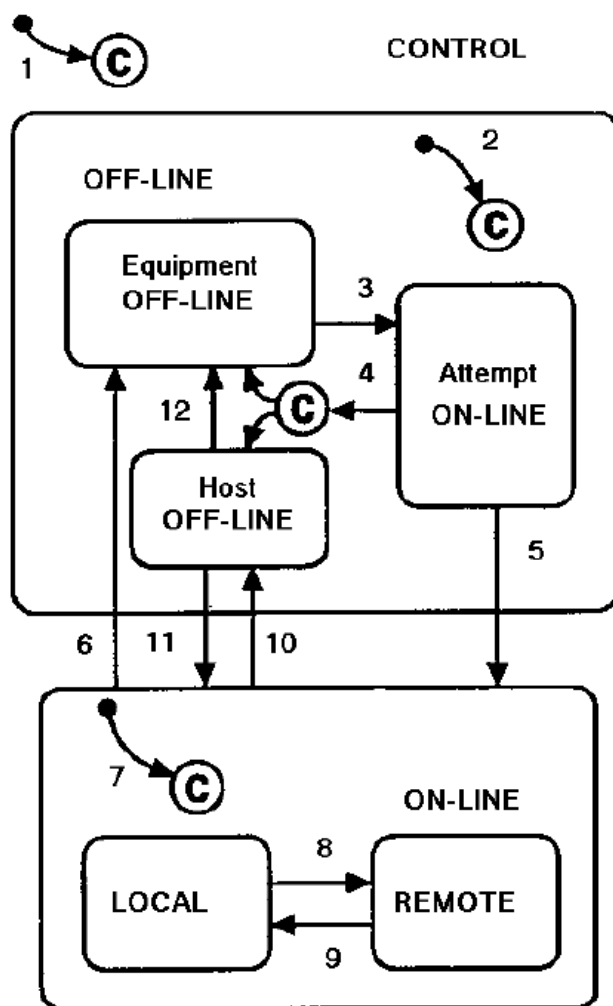
The CONTROL state model defines the level of cooperation between the host and equipment.

OFFLINE : When the OFF-LINE state is active, operation of the equipment is performed by the operator at the operator console. While the equipment is OFF-LINE, message transfer is possible. However the use of messaging for any automation purpose is severely restricted. While the OFF-LINE state is active, the equipment will only respond to those messages used for the establishment of communications or a host request to activate the ONLINE state. While OFF-LINE, the equipment will respond with an Sx,F0 to any primary message from the host other than S1,F13 or S1,F17. It will process and respond to S1,F13 and S1,F17. S1,F17 is used by the host to request the equipment to transition to the ON-LINE state. The equipment will accept this request and send a positive response only when the HOST OFF-LINE state is active (see transition 11 definition below).

While the OFF-LINE state is active, the equipment shall attempt to send no primary message other than S1,F13, S9,Fx, and S1,F1 (see ATTEMPT ONLINE substate). If the equipment receives a reply message from the host other than S1,F14 or S1,F2, this message is discarded. No messages enter the spool when the system is OFFLINE. Spooling may be active when the Communications State of NOT COMMUNICATING is active. This might occur during OFF-LINE, but since the equipment will not attempt to send messages except as mentioned in the previous paragraph, no messages will enter the spool.

OFF-LINE has three substates: EQUIPMENT OFFLINE, ATTEMPT ON-LINE, and HOST OFF-LINE. OFF-LINE/EQUIPMENT OFF-LINE While this state is active, the system maintains the OFFLINE state. It awaits operator instructions to attempt to go ON-LINE. OFF-LINE/ATTEMPT ON-LINE While the ATTEMPT ON-LINE state is active, the equipment has responded to an operator instruction to attempt to go to the ON-LINE state. Upon activating this state, the equipment attempts to send an S1,F1 to the host

ON-LINE : While the ON-LINE state is active, SECS-II messages may be exchanged and acted upon. Capabilities that may be available to the host should be similar to those available from the operator console wherever practical. The use of Sx,F0 messages is not required while the ON-LINE state is active. Their use is discouraged in this case. The only allowed use is to close open transactions in conjunction with message faults.



Status	Description
EQUIPMENT OFF-LINE (CEID 1)	While this state is active, the system maintains the OFFLINE state. It awaits operator instructions to attempt to go ON-LINE.
ATTEMPT ON-LINE	While the ATTEMPT ON-LINE state is active, the equipment has responded to an operator instruction to attempt to go to the ON-LINE state. Upon activating this state, the equipment attempts to send an S1,F1 to the host. Note that when this state is active, the system does not respond to operator actuation of either the ON-LINE or the OFF-LINE switch.
HOST OFF-LINE (CEID 1)	While the HOST OFF-LINE state is active, the operator's intent is that the equipment be ON-LINE. However, the host has not agreed. Entry to this state may be due to a failed attempt to go ON-LINE or to the host's request that the equipment go OFF-LINE from ON-LINE (see the transition table for more detail). While this state is active, the equipment shall positively respond to any host's request to go ON-LINE (S1,F17). Such a request shall be denied when the HOST OFFLINE state is not

LOCAL (CEID 2)	active. Operation of the equipment is implemented by direct action of an operator. All operation commands shall be available for input at the local operator console of the equipment. The host shall have the following capabilities and restrictions when the LOCAL state is active: — The host shall be prohibited from the use of remote commands that cause physical movement or which initiate processing. During processing, the host shall be prohibited from the use of any remote command that affects that process. — During processing, the host shall be prohibited from modifying any equipment constants that affect that process. Other equipment constants shall be changeable during processing. The host shall be able to modify all available equipment constants when no processing is in progress. — The host shall be capable of initiating the upload and download of recipes to/from the recipe storage area on the equipment. The host shall be capable of selecting recipes for execution so long as this action does not affect any currently executing recipe. — The host shall be able to configure automatic data reporting capabilities including alarms, event reporting, and trace data reporting. The host shall receive all such reports at the appropriate times. — The host shall be able to inquire for data from the equipment, including status data, equipment constants, event reports, process program directories, and alarms. — The equipment shall be able to perform Terminal Services as defined in GEM. The host shall be allowed any other capabilities that were not specifically restricted in the above items as long as the LOCAL state is active.
REMOTE (CEID 3)	For equipment which supports the GEM capability of remote control , while the REMOTE state is active, the host shall have access, through the communications interface, to the necessary commands to operate the equipment through the full process cycle in an automated manner. The equipment does not restrict any host capabilities when REMOTE is active. The degree of control executed by the host may vary from factory to factory. In some cases, the operator maybe required to interact during remotely controlled processes. This interaction may involve set-up operations, operator assist situations, and others. This state is intended to be flexible enough to accommodate these different situations. To support the different factory automation policies and procedures, it shall be possible to configure the equipment to restrict the operator in specific nonemergency procedures. These restrictions shall be configurable so that the equipment may be set up to allow the operator to perform

necessary functions without contention with the host. The categories for configuration shall include (but are not limited to):

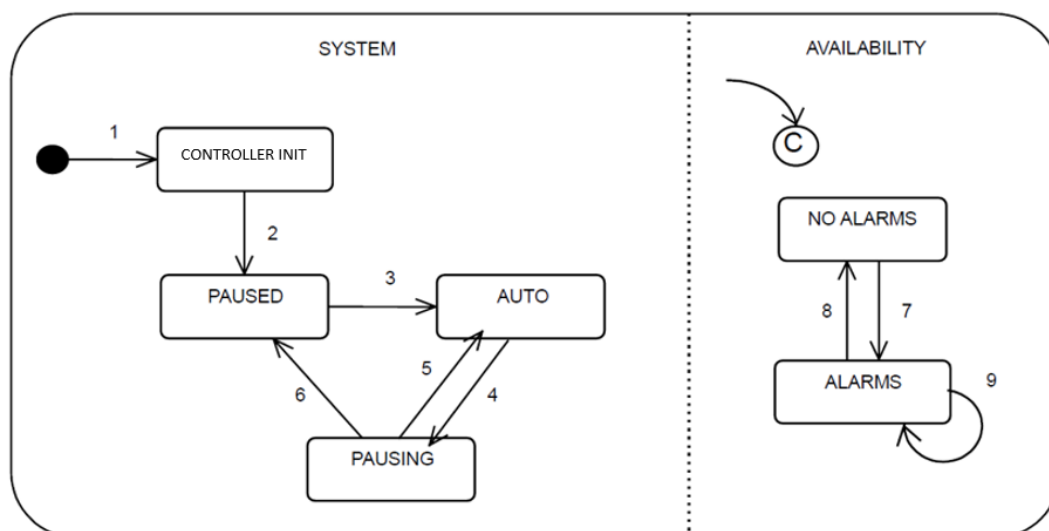
- change equipment constants (process-related),
- change equipment constants (non-process-related),
- initiate process program download,
- select process program,
- start process program,
- pause/resume process program,
- operator assist,
- material movement to/from equipment,
- equipment-specific commands (on a command-by command basis if needed).

02.02 Status Models

02.02.01 Controller Status Definitions

The purpose of the Controller System state model is to provide information to the host regarding the overall status of the Controller System. The Controller state model is valid when the SEMI E30 (GEM) state is ON-LINE. The Controller state model is not valid when the SEMI E30 (GEM) state is OFF-LINE. Since a controller system may consist of many components (e.g., vehicle, robot arm, ID reader, etc.), it may be possible to continue ON-LINE operation when the operation mode of some controller components (as viewed by the Controller) is a manual state. The details of what happens when individual components of the controller system enter a

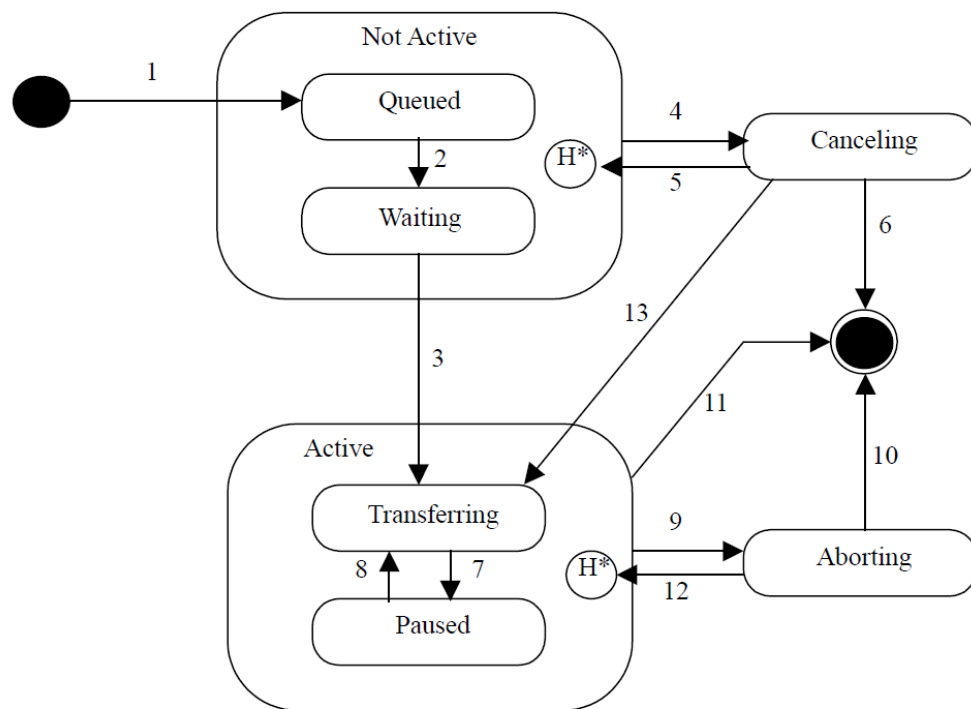
manual state are specific to the IBSEM equipment supplier. When the SEMI E30 Control state changes from OFF-LINE to ON-LINE, the Controller State Model is started from the Controller INIT state.



Status	Description
INIT (CEID 54)	Controller initialization of Controller components is occurring. This is a non-operational state. No commands from the host will be processed or queued. The system will not move out of this state if there are vehicles actively loading or unloading carriers at ports. These vehicles must be manually or automatically recovered before moving on to the next state
PAUSING (CEID 57)	A system PAUSE command has been received and is being processed. All vehicles that are currently loading or unloading will continue until the load/unload is complete. Vehicles that are currently moving may continue to move but they must not begin a load or unload. TRANSFER commands are accepted and queued. All status requests will be processed. The RESUME command will also be processed.
PAUSED (CEID 55)	No vehicles are in the process of loading or unloading a carrier at a port, but vehicles may still be moving. TRANSFER commands are accepted and queued. All status requests will be processed. The RESUME command will also be processed.
AUTO (CEID 53)	System is in the normal operational state. Commands are actively processed.
NO ALARMS	There are no alarms present in the system.
ALARMS	There are one or more alarms present in the system, but the Controller is still capable of normal processing since several components may remain unaffected by the alarm situation.

02.02.02 TRANSFER Command State Model

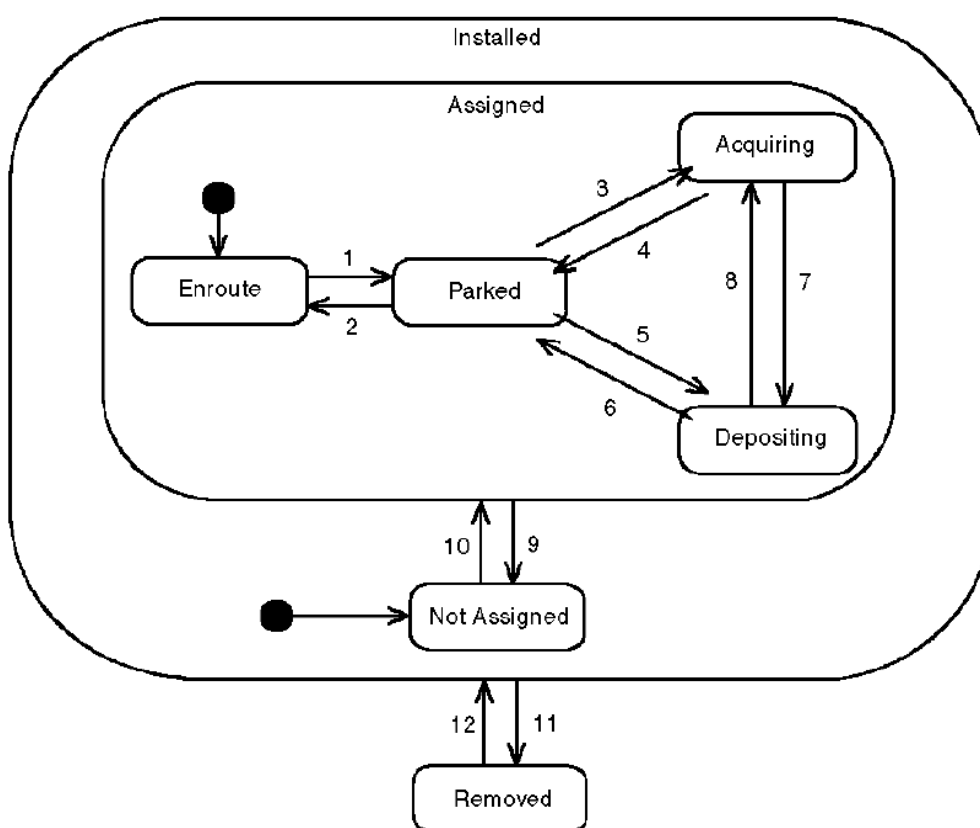
The TRANSFER command state model serves as the SEMI E30 Processing State Model. The purpose of the TRANSFER command state model is to provide information to the host regarding the control of the TRANSFER command. The TRANSFER command allows the host to manage interbay or intrabay delivery and scheduling. The control of each TRANSFER command must independently support the TRANSFER command state model.



Status	Description
NOT ACTIVE	The transfer unit is not involved in the physical aspect of the TRANSFER command. It is denoted by the time spanned by the queuing of the TRANSFER command to the moment just prior to the acquire of the first carrier in the transfer unit.
QUEUED (NOT ACTIVE sub-state)	Controller has acknowledged and queued the TRANSFER command. TRANSFER command has not been initiated.
WAITING (NOT ACTIVE sub-state)	TRANSFER command has been initiated.
ACTIVE	The transfer unit is involved in the physical aspect of the TRANSFER command. It is denoted by the time spanned by the acquire of the first carrier in the transfer unit to the deposit of the last carrier in the transfer unit.
TRANSFERRING (ACTIVE sub-state)	transfer command is actively being executed by the transport equipment.
PAUSED (ACTIVE sub-state)	The transfer command is not actively being executed by the transport equipment.
CANCELING	The TRANSFER command cancel procedure is being performed to terminate a transfer command which never entered the ACTIVE state (either QUEUED or WAITING). This state is entered via a CANCEL remote command. An unsuccessful transfer command completion will ultimately result.
ABORTING	The TRANSFER command abort procedure is being performed to terminate a transfer command which has entered the ACTIVE state. This state can only be entered via an ABORT remote command. An unsuccessful transfer command completion will ultimately result from this state.

02.02.03 Vehicle State Model

The purpose of the vehicle state model is to provide information to the host for use of transport vehicle information and metric tracking (i.e., the Host will not control vehicles). Each vehicle must individually comply with the vehicle state model. Implementation of this state model is not a requirement for Transport Systems which do not have vehicles. An example of such a Transport System is a DWC. The Host should not be dependent on any events from the vehicle state model.



Status	Description
INSTALLED (CEID 206)	The vehicle is available or being used for TRANSFER commands. All enabled collection events and alarms will be sent to the Host for vehicles in this state.
REMOVED (CEID 205)	The vehicle is not available for Host initiated TRANSFER commands. No collection events or alarms will be sent to the Host for vehicles in this state.
ASSIGNED (INSTALLED sub-state) (CEID 220)	Vehicle is allocated to a TRANSFER command.
NOT ASSIGNED (INSTALLED sub-state) (CEID 221)	Vehicle is not allocated to a TRANSFER command.

ENROUTE (ASSIGNED sub-state)	The vehicle is on its way to a transfer port. This is the default entry into the ASSIGNED state since it must be entered for the host to track vehicle metrics completely and adequately.
PARKED (ASSIGNED sub-state)	This state occurs when the vehicle is in the following conditions: • After the arrival of the vehicle is completed and before the action of the transfer agent is started. • After the action of the transfer agent has completed and before the departure of the vehicle. • After continuous actions of the transfer agent (e.g., acquire/acquire and deposit/deposit) are completed.
ACQUIRING (ASSIGNED sub-state) (CEID 210)	The vehicle is currently involved in carrier acquire (one or more carriers possible depending on vehicle limitations).
DEPOSITING (ASSIGNED sub-state) (CEID 212)	The vehicle is currently involved in carrier deposit (one or more carriers possible depending on vehicle limitations).

02.02.04 Carrier State Model

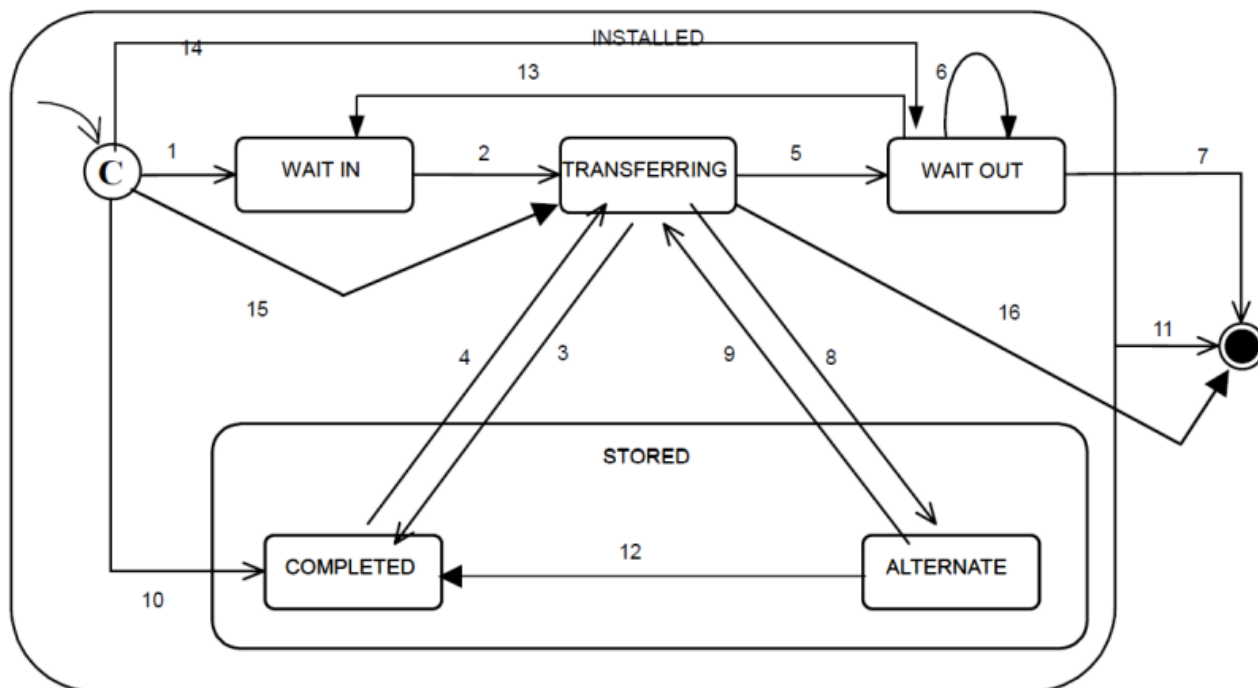
The purpose of the carrier state model is to provide information to the host regarding carrier tracking (the Host will not control carriers). Each carrier must comply with the carrier state model.

02.02.04.01 IB SEM



Status	Description
INSTALLED (CEID 151)	A Controller database entry exists for the carrier.

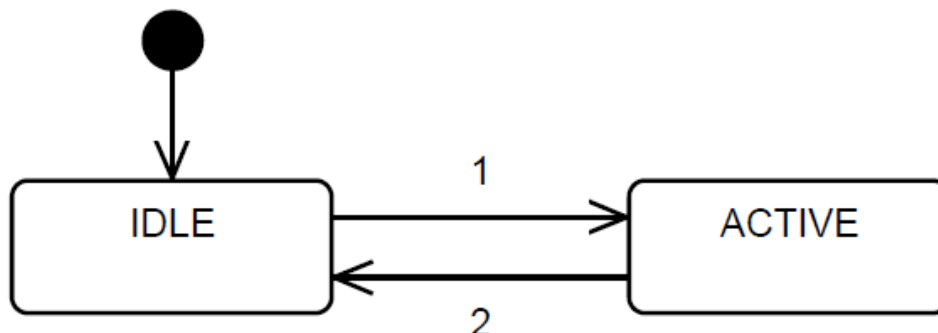
02.02.04.02 Stock SEM



Status	Description
INSTALLED (CEID 151)	A Controller database entry exists for the carrier.
WAIT IN(INSTALLED sub-state) (CEID 158)	Carrier at an internal input location in the stocker ready to be moved to a storage or output location
WAIT OUT(INSTALLED sub-state) (CEID 159)	Carrier at an output location ready to be removed manually or automatically
TRANSFERRING(INSTALLED sub-state) (CEID 160)	Carrier moving between locations in the stocker
STORED(INSTALLED sub-state)	Carrier is sitting at a storage location.
COMPLETED(STORED sub-state)	The carrier is stored at a shelf location as a result of completing a transfer for which this shelf represents the destination of the transfer command.
ALTERNATE(STORED sub-state)	The Carrier is temporarily stored at a shelf location. The transfer command completion is pending until the destination becomes available.
BLOCKED(STORED sub-state)	As for the carrier, it was at the position of the shelf, and was done the transportation prohibition.

02.02.05 Stocker Crane State Model

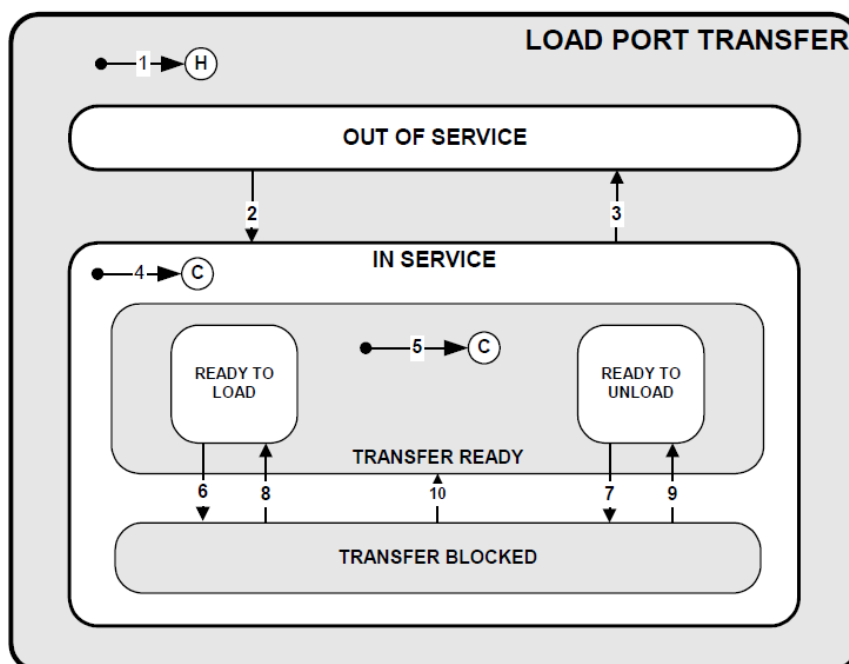
The purpose of the stocker crane state model is to provide information to the host for use of stocker crane information and metric tracking



Status	Description
IDLE(CEID 202)	The stocker crane is not performing Host or SC initiated work
ACTIVE(CEID 201)	The stocker crane is busy performing Host or SC initiated work.

02.02.06 Port State Model

A load port (port) is used by the factory to load and unload carriers to and from production equipment. A load port may be used as an input load port, an output load port, or as an input/output load port, depending upon equipment type, configuration and/or factory practices. This classification may be fixed or it may be programmable by the user. A load port is generally designed to handle one specific carrier type, such as substrate cassettes, leadframe magazines, SMIF pods, or FOUPs.



Status	Description
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OUT OF SERVICE (CEID 701)	Transfer to/from this load port is disabled. A transition to IN SERVICE is required to continue using this load port for transfers.
IN SERVICE (CEID 702)	Transfer to/from this load port is enabled. A transition to OUT OF SERVICE disables the load port for transfer use.
READY TO LOAD	A sub-state of TRANSFER READY. When transitioning to the TRANSFER READY state, if a carrier is not present on the specified load port, this is the active sub-state. In this state, the load port is available to be loaded with an external carrier, or with a carrier that is currently located inside the equipment (i.e. internal buffer).
READY TO UNLOAD	A sub-state of TRANSFER READY. When transitioning to the TRANSFER READY state, if a carrier is present on the specified load port, this is the active sub-state. In this state, the load port is available for unloading of a carrier from the loadport to material handling equipment. When the load port is being used by the equipment, the state shall transition to TRANSFER BLOCKED.
TRANSFER BLOCKED	The carrier transfer state is neither READY TO LOAD nor READY TO UNLOAD. Because of load port related activity being performed, transfer is not available to/from this load port at this time.

3. 基本功能需求(Common Function Requirements)

03.01 設備功能(Equipment Function)

03.01.01 通信線路檢查功能 Communication Line Check Function

The objective of this function is to check the reliability of communication between Equipment and Host. When Equipment receives loop-back diagnostic request from Host, it should reply back to the host with the same string. Also, Host can perform the same action to Equipment when operating in any time (no matter On Line and Off Line Mode).

該功能的目的是檢查設備和主機之間通信的可靠性。當設備收到來自主機的環回診斷請求時，它應該使用相同的字串回復主機。此外，主機可以在任何時間（無論是在線和離線模式）操作時對設備執行相同的操作。

03.01.02 狀態請求功能(Status Request Function)

Host can sent the status query request to Equipment at any time in On Line status.

The every status query includes the following :

(1). Equipment status query,

(2). Communication mode status query,

(3). Material list or status query.

主機可以在線狀態隨時向設備發送狀態查詢請求。

每個狀態查詢包括以下內容：

(1) 設備狀態查詢，

(2) 通訊模式狀態查詢，

(3) 材料清單或狀態查詢。

03.01.03 通信模式狀態變更報告功能(Communication Mode Status Change Report Function)

(1) After operator switches from the Off Line Mode to On Line Mode at the Equipment, it should send the status report to Host accordingly when the communication has been established.

(2) When operator switches from On Line Mode to Off Line Mode at Equipment side, Equipment should send the corresponding mode change report to Host.

(3) When Off Line Mode is being switched to On Line but Equipment didn't receive the acknowledgement from Host or Host reply reject, the Equipment should remain at the original state (Off Line).

(1) 操作員在設備處從離線模式切換到在線模式後，應在通信建立後相應地將狀態報告發送給主機。

(2) 當操作員在設備側從在線模式切換到離線模式時，設備應向主機發送相應的模式更改報告。

(3) 當離線模式切換到聯機狀態但設備未收到來自主機或主機回復拒絕的確認時，設備應保持原始狀態（離線）。

03.01.04 狀態改變報告功能(Status Change Report Function)

(1) When equipment status (refer to section 04.02 Equipment Status Definition) changes, it has to report status change event to Host under On Line Mode.

(2) When material status (refer to 04.03 Material Status Definition) changes, it has to report status change event to Host under On Line Mode.

(1) 當設備狀態（參見 04.02 設備狀態定義）發生變化時，必須在聯機模式下向主機報告狀態變化事件。

(2) 當材料狀態（參見 04.03 材料狀態定義）發生變化時，它必須在聯機模式下向主機報告狀態變化事件。

03.01.05 報警事件報告功能(Alarm Event Report Function)

When Equipment have any alarm happened, equipment should sent alarm report to Host (No matter Off Line/On Line Mode when the HSMS communication is connected). The alarm

report items including alarm level and alarm event, alarm code and alarm text that affected by the alarm. Also, when alarm already is cleared, equipment should sent alarm clear event and report to Host

當設備發生任何報警時，設備應向主機發送報警報告（無論是 HSMS 通信連接時的離線/在線模式）。報警報告項目包括報警級別和報警事件，報警代碼和受報警影響的報警文本。此外，當警報已經清除時，設備應發送警報清除事件並向主機報告

03.01.06 SECS 錯誤報告事件(SECS Error Report Function)

Equipment sent SECS error report to Host, when the block message cannot be handled or timeout error cause transaction failure. (Refer to S9Fx)

設備向 Host 發送 SECS 錯誤報告，當無法處理阻塞信息或超時錯誤導致事務失敗。（參考 S9Fx）

4. SECS Stream 功能(SECS Stream Function)

04.01 Stream Function List

Host → Equipment (Primary)		Equipment → Host (Secondary)	
S1F1	Are You There Request	S1F2	On-Line Data
S1F3	Select Equipment Status Request	S1F4	Select Equipment Status Data
S1F11	Status Variable Name List Request	S1F12	Status Variable Name List Reply
S1F13	Establish Communications Request	S1F14	Establish Communications Request Acknowledge
S1F15	Request Off-Line	S1F16	Off-Line Acknowledge
S1F17	Request On-Line	S1F18	On-Line Acknowledge
S2F13	Equipment Constant Request	S2F14	Equipment Constant Data
S2F15	New Equipment Constant Send	S2F16	New Equipment Constant Acknowledge
S2F17	Date Time Request	S2F18	Date Time Reply
S2F25	Loop-back Diagnostic Request	S2F26	Loop-back Diagnostic Data
S2F29	Equipment Constant Namelist Request	S2F30	Equipment Constant Namelist
S2F31	Date and Time Send	S2F32	Date and Time Acknowledge
S2F33	Define Report	S2F34	Define Report Acknowledge
S2F35	Link Event Report	S2F36	Link Event Report Acknowledge
S2F37	Enable/Disable Event Report	S2F38	Enable/Disable Event Report Acknowledge
S2F41	Host Command Send	S2F42	Host Command Acknowledge
S2F49	Enhanced Remote Command	S2F50	Enhanced Remote Command Acknowledge
S5F3	Enable/ Disable Alarm Send	S5F4	Enable/ Disable Alarm Acknowledge

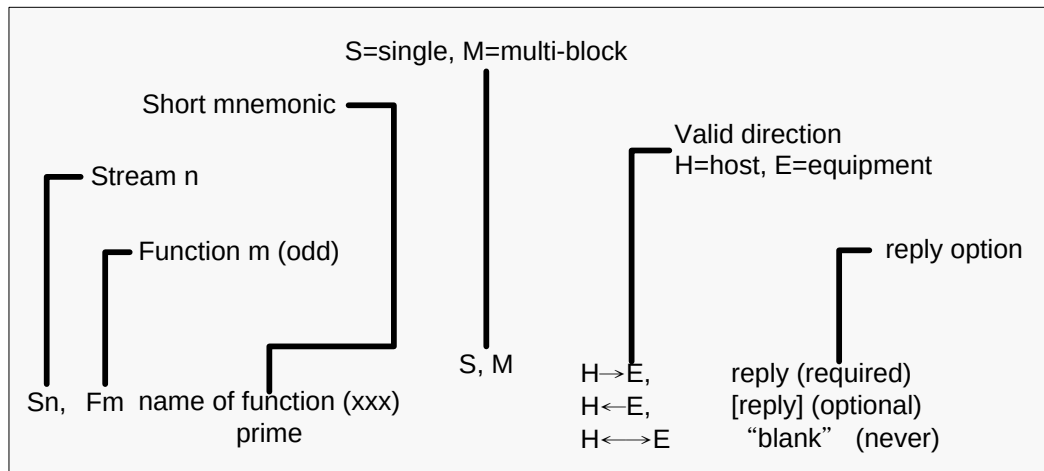
Host → Equipment (Secondary)		Equipment → Host (Primary)	
SxFx		SxF0	Abort Transaction
S1F2	On-Line Data	S1F1	Are You There Request
S1F14	Establish Communications Request Ack.	S1F13	Establish Communications Request
S2F18	Date Time Reply	S2F17	Date Time Request
S2F26	Loop-back Diagnostic Data	S2F25	Loop-back Diagnostic Request
S5F2	Alarm Report Acknowledge	S5F1	Alarm Report Send
S6F12	Event Report Acknowledge	S6F11	Event Report Send
		S9F1	Unrecognized Device ID
		S9F3	Unrecognized Stream Type

		S9F5	Unrecognized Function Type
		S9F7	Illegal Data
		S9F9	Transaction Timer Timeout
		S9F11	Data Too Long
		S9F13	Conversation Timeout

04.02 Stream 功能描述(Stream Function Description)

The sections are described in a standard form which involves specification of the number, name, single or multiple blocks, direction of communication, nature of reply required, description, variable definition, and the detailed structure of the message in terms of lists and items. Double lines separate streams, and single lines separate transactions to aid readability. The abbreviations used in each transaction are as follows:

這些部分以標準形式描述，其涉及數量，名稱，單個或多個塊的指定，通信方向，所需答复的性質，描述，變量定義以及根據列表和項目的消息的詳細結構。雙行分隔流，單行分隔事務以提高可讀性。每筆交易使用的縮寫如下：



Description: A description of the action generated by the function.

Item Definition: It's definition of each item by the function.

Structure: Detailed structure showing lists and defined items. Lists are denoted by a capital L followed by the length separated by a comma. The individual elements in the list are numbered on separate lines. Nested lists are indented to emphasize the structure. The detailed form of the items is given in the define section at the beginning of the transaction. The symbols "<" and ">" are used to enclose each item in the structure data and imply that there is an item header. A detailed description of each data item as well as a list of the allowable data formats can be found in the item description.

Exception: special cases in the structure that have a different meaning.

描述：函數生成的操作的描述。

項目定義：由功能定義每個項目。

結構：顯示列表和已定義項目的詳細結構。列表由大寫字母 L 表示，後跟由逗號分隔的長度。列表中的各個元素在不同的行上編號。嵌套列表縮進以強調結構。項目的詳細形式在交易開始時的定義部分給出。符號 "<" 和 ">" 用於包含結構數據中的每個項目，並暗示存在項目標題。可以在項目描述中找到每個數據項的詳細描述以及允許的數據格式列表。

例外：結構中具有不同含義的特殊情況。

04.02.01 SnF0 Abort Transaction**Sn, F0 Abort Transaction****S,H←E.**

Description: Used in lieu of an expected reply to abort a transaction. Stream n and Function 0 are defined in every stream and has the same meaning in every stream.

Structure: Header only.

04.02.02 S1F1 Are You There Request**S1, F1 Are You There Request****S,H↔E, reply.**

Description: Establishes if the equipment is on-line. A function 0 response to this message means the communication is inoperative. In the equipment, a function 0 is equivalent to a timeout on the receive timer after issuing S1, F1 to the host.

Structure: Header only.

04.02.03 S1F2 On Line Data**S1, F2 On Line Data****S,H↔E.**

Description: Data signifying that equipment is alive.

Structure:

L, 2

1. <MDLN>
2. <SOFTREV>

Exception: The host sends a zero-length list to the equipment.

04.02.04 S1F3 Selected Equipment Status Request**S1, F3 Selected Equipment Status Request****S,H→E, reply.**

Description: A request to the equipment to report selected values of its status.

Structure:

L,n

- 1.<SVID>
- ...
- n.<SVID>

Exception: A zero-length list (structure 1) or item (structure 2) means report all SVIDs.

Note: At the time when the Item length is 0 (n=0), all SVID Report is requested.

04.02.05 S1F4 Selected Equipment Status Reply**S1, F3 Selected Equipment Status Reply****S,H←E**

Description: The equipment reports the value of each SVID requested in the order requested.

The host remembers the names of values requested.

Structure:

L,n
1.<SV>
...
n.<SV>

Exception: A zero-length list item for SV_i means that SVID_i does not exist

Note: If the requested SVID is illegal, it sends "List 0".

04.02.06 S1F11 Status Variable Name List Request**S1, F5 Status Variable Name List Request****S,H→E, reply.**

Description: A request to identify certain status variables

Structure:

L,n
1.<SVID>
...
n.<SVID>

Exception: A zero-length means report all SVIDs.

04.02.07 S1F12 Status Variable Name List Reply**S1, F6 Status Variable Name List Reply****M,H←E.**

Description: Reports to the host the name and unit of the requested SVs

Structure:

1.L,n
1.L.3
1.<SVID1>
2.<SVNAME1>
3.<UNITS>

- ...
- n.L.3
- 1.<SVIDn>
 - 2.<SVNAME_n>
 - 3.<UNITS>

Exception: A zero-length means report all SVIDs.

04.02.08 S1F13 Establish Communication Request

S1, F13 Establish Communication Request

S,H $\longleftrightarrow$ E, Reply.

Description: The purpose of this message is to provide a formal means of initializing communications at a logical application level both on power-up and following a break in communications. It should be the following any period where host and equipment SECS applications are unable to communicate. An attempt to send an Establish Communications Request (S1F13) should be repeated at programmable intervals until an Establish Communications Acknowledge (S1F14) is received within the transaction timeout period with an acknowledgement code accepting the establishment.

Item Definition: None.

Structure:

- L,2
- 1.<MDLN>
 - 2.< SOFTREV>

Exception: The host sends a zero-length list to the equipment.

Note : Host will return "List 0" to the equipment.

04.02.09 S1F14 Establish Communication Request Acknowledge

S1, F14 Establish Communication Request Acknowledge

S,H $\longleftrightarrow$ E.

Description: Accept or deny Establish Communications Request (S1F13).MDLN and SOFTREV are on-line data and are valid only if COMMACK = 0.

Structure:

- L,2
- 1.<COMMACK>
 2. L,2
 - 1.<MDLN>
 - 2.< SOFTREV >

Exception: The host sends a zero-length list for item 2 to the equipment.

Note: Host will return "List 0" for item 2 to the equipment.

04.02.10 S1F15 Off Line Request

S1, F15 Off Line Request

S,H→E, Reply.

Description: The Host requests that the equipment transition to the OFF-Line state.

Structure: Header only.

04.02.11 S1F16 Off Line Acknowledge

S1, F16 Off Line Acknowledge

S,H←E.

Description: Acknowledge or error in response to S1, F15.

Structure:

<OFLACK>

04.02.12 S1F17 On Line Request

S1, F17 On Line Request

S,H→E, Reply.

Description: The Host requests that the equipment transition to the ON-Line state.

Structure: Header only.

04.02.13 S1F18 On Line Acknowledge

S1, F18 On Line Acknowledge

S,H←E.

Description: Acknowledge or error in response to S1, F17.

Structure:

<ONLACK>

04.02.14 S2F13 Equipment Constant Request

S2, F13 Equipment Constant Request

S,H→E, Reply.

Description: Host sends L:0 to receive all in predefined order.

Structure:

<ECID>

04.02.15 S2F14 Equipment Constant Data**S2, F14 Equipment Constant Data****S,H←E.**

Description: An L:0 is substituted if an ECID does not exist, L format is not allowed for an ECV.

Structure:

<ECV>

04.02.16 S2F15 New Equipment Constant Send**S2, F15 New Equipment Constant Send****S,H→E, Reply.**

Description: Change one or more equipment constants. The only constant set by the MCS for the Stocker Controller will be the device name.

Structure:

L,n

1.L,2

1.<ECID1>

2.<ECV1>

:

n.L,2

1.<ECIDn>

2.<ECVn>

04.02.17 S2F16 New Equipment Constant Acknowledge**S2, F16 New Equipment Constant Acknowledge****S,H←E.**

Description: Acknowledge or error. If EAC contains a non-zero error code, the equipment should not change any of the ECIDs specified in S2F15.

Structure:

<ECV>

04.02.18 S2F17 Date and Time Request**S2, F17 Date and Time Request****S,H↔E, reply.**

Description: Useful to check equipment time base or for equipment to synchronize with the host time base.

Structure: Header only.

04.02.19 S2F18 Date and Time Data**S2, F18 Date and Time Data****S,H $\longleftrightarrow$ E.**

Description: Acknowledge the receipt of time and date.

Structure:

<DATETIME> YYYYMMDDHHMMSS

04.02.20 S2F25 Loop-back Diagnostic Request**S2, F25 Loop-back diagnostic Request****S,H $\longleftrightarrow$ E, reply.**

Description: A diagnostic message for checkout of protocol and communication circuits. The binary string sent is echoed back.

Structure:

<ABS> Any Binary Data

04.02.21 S2F26 Loop-back Diagnostic Data**S2, F26 Loop-back Diagnostic Data****S,H $\longleftrightarrow$ E.**

Description: The echoed binary string.

Structure:

<ABS> Any Binary Data

04.02.22 S2F29 Equipment Constant Name list Request**S2, F29 Equipment Constant Name list Request****S,H $\rightarrow$ E, reply.**

Description: The function allows the host to retrieve basic information about what equipment constants are available.

Structure:

1.L,n

1.<ECID1>

2.<ECID2>

...

n.<ECIDn>

Exception: A zero-length list send information for all ECIDs.

04.02.23 S2F30 Equipment Constant Namelist**S2, F30 Equipment Constant Namelist****S,H $\leftarrow$ E.**

Description: Data response

Structure:

1.L,n

1.L.6

1. <ECID 1>
2. <ECNAME 1>
3. <ECMIN 1>
4. <ECMAX 1>
5. <ECDEF 1>
6. <UNITS 1>

...

n.L.6

1. <ECID n>
2. <ECNAME n>
3. <ECMIN n>
4. <ECMAX n>
5. <ECDEF n>
6. <UNITS n>

04.02.24 S2F31 Date and Time Set Request

S2, F31 Date and Time Set Request

S,H→E, reply.

Description: Useful to synchronize the equipment time with the host time base.

Structure:

<DATETIME> "YYYYMMDDhhmmss"

04.02.25 S2F32 Date and Time Set Acknowledge

S2, F32 Date and Time Set Acknowledge

S,H←E.

Description: Useful to synchronize the equipment time with the host time base.

Structure:

<TIACK>

04.02.26 S2F33 Define Report

S2, F33 Define Report

S,H→E, reply.

Description: The purpose of this message is for the host to define a group of reports for the Stocker Controller. Although the SEMI standard supports multi-block *Define Report*

messages, only single block messages are needed for this interface specification.

Structure:

```

L,2
  1.<DATAID> :
  2. L,a # of Reports
    1.L,2
      1.<RPTID1> /* Report #1*/
      2.L,b
        1.<VID1>
        :
        b.<VIDb>
      :
    a.L,2
      1.<RPTIDa> /* Report #a*/
      2.L,b
        1.<VID1>
        :
        b.<VIDb>

```

Exception:

1. A list of zero-length following <DATAID> deletes all report definitions and associated links. (See S2F35 – Link Event Report)
2. A list of zero-length following <RPTID> deletes report type, RPTID. All CEID links to this RPTID are also deleted.

Note:

A list of 0 length following <DATAID> deletes all report definitions and associated links. A list of 0 length following <RPTID>delete Report Type <RPTID>. And all CEID Link to this RPTID is also deleted.

04.02.27 S2F34 Define Report Acknowledge

S2, F34 Define Report Acknowledge

S,H←E.

Description: Acknowledge or error. If an error condition is detected, the entire message is rejected. (i.e., partial changes are not allowed.).

Structure:

< DRACK >

04.02.28 S2F35 Link Event Report**S2, F35 Link Event Report****S,H→E, reply.**

Description: The purpose of this message is for the host to link n reports to an event (CEID).

These event reports will default to 'disabled' upon linking. That is, the occurrence of an event would not cause the report to be sent until enabled. (See S2F37 – Enable/Disable Event Report for enabling reports.).

Structure:

```
L,2
  1.<DATAID> :
  2. L,a # of Events
    1.L,2
      1.<CEID1> /* Event #1*/
      2.L,b
        1.<RPTID1>
        :
        b.<RPTIDb>
      :
    a.L,2
      1.<CEIDa> /* Report #a*/
      2.L,b
        1.<RPTID1>
        :
        b.<RPTIDb>
```

Exception: A list of zero-length following <CEID> deletes all report links to that event.

Note:

1. Although the SEMI standard supports multi-block Define Report messages, only single block messages are needed for this interface specification.
2. The MCS will only define one report per CEID.
3. A list of 0 length following CEID deletes all reports linked to that event.

04.02.29 S2F36 Link Event Report Acknowledge**S2, F36 Link Event Report Acknowledge****S,H←E.**

Description: Acknowledge or error. If an error condition is detected, the entire message is rejected. (i.e., partial changes are not allowed.).

Structure:

< LRACK >

04.02.30 S2F37 Enable/Disable Event Report**S2, F37 Enable/Disable Event Report****S,H→E, reply.**

Description: The purpose of this message is for the host to enable or disable reporting for a group of events (CEIDs).

Structure:

- L,2
 - 1.<CEED> Enable / Disable
 - 2. L,n # of CEIDs
 - 1.<CEID1>
 - :
 - n.<CEIDn>

Exception: A list of zero-length following <CEED> means the enable or disable action should be taken for all CEIDs.

04.02.31 S2F38 Enable/disable Event Report Acknowledge**S2, F38 Enable/disable Event Report Acknowledge****S,H←E.**

Description: Acknowledge or error. If an error condition is detected, the entire message is rejected. (i.e., partial changes are not allowed.).

Structure:

< ERACK >

04.02.32 S2F41 Host Command Send**S2, F41 Host Command Send****S,H→E, reply.**

Description: The host requests the equipment to perform the specified remote command with the specified parameters.

Structure:

- L,2
 - 1.<RCMD>
 - 2.L,n #of Parameters
 - 1.L,2
 - 1.<CPNAME1>
 - 2.<CPVAL1>
 - :
 - n.L,2

1.<CPNAME_n>2.<CPVAL_n>**04.02.33 S2F42 Host Command Acknowledge****S2, F42 Host Command Acknowledge****S,H←E.**

Description: Acknowledge host command or error. I command is not accepted due to one or more invalid parameters, (i.e., HCK=3), then a list of invalid parameters will be returned containing the parameter name and reason for being invalid.

Structure:

L,2

1.<HCK>

2.L,n #of Parameter

1.L,2

1.<CPNAME1>

2.<CPACK1>

:

n.L,2

1.<CPNAME_n>2.<CPACK_n>

Exception:

If there are no invalid parameters, then a list of zero length will be sent for item 2.

04.02.34 S2F49 Enhanced Remote Command Extension**S2, F49 Enhanced Remote Command Extension****S,H→E, reply.**

Description: The host requests the equipment to perform the specified remote command with the specified parameters.

Structure:

L,4

1.<DATAID>

2. <OBSPEC>

3.<RCMD>

4.L,m #of Parameter Groups

1.L,2

1.<CPNAME1> #Command Parameter 1 Name

2.<CEPVAL1> #Cmd-Enhanced Parameter 1 Value

:

m.L,2

1.<CPNAMEm> #Command Parameter m Name

2.<CEPVALm> #Cmd-Enhanced Parameter m Value

Exception:

A zero length LIST (m=0) indicates that no parameter groups are sent with the command.

OBJSPEC can be a null length item

Note: If the specific value of CPNAME is defined to have a CEPVAL defined as a LIST, it shall always be a LIST. If CEPVAL that is associated to that specific value of CPNAME is defined to be anything other than LIST, it will result in a format error.

TRANSFER:

'S2F49' W /* Enhanced Remote Command H->E */

< L[4]

< U4[1] 1 > *DATAID

< A[0] " > *OBJSPEC

< A[8] 'TRANSFER' > *RCMD

< L[3]

< L[2]

< A[11] 'COMMANDINFO' > *CPNAME

< L[2] *CEPVAL

< L[2]

< A[9] 'COMMANDID' > *COMMANDID

< A[6] '2021011814373154' > *COMMANDID value

>

< L[2]

< A[8] 'PRIORITY' > *PRIORITY

< U2[1] 5 > *PRIORITY value

>

>

>

< L[2]

< A[12] 'TRANSFERINFO' > *CPNAME

< L[4] *CEPVAL

< L[2]

< A[9] 'CARRIERID' > *CARRIERID

< A[10] 'FJ09D00111' > *CARRIERID value

>

< L[2]

```

        < A[6] 'SOURCE' > * SOURCE
        < A[6] 'NormalZone_1' > * SOURCE value
    >
    < L[2]
        < A[4] 'DEST' > * DEST
        < A[6] 'Port_1_1' > * DEST value
    >
    < L[2]
        < A[4] 'NEXTDEST' > * NEXTDEST
        < A[6] 'Port_1_2' > * NEXTDEST value
    >
    >
    >
    < L[2]
        < A[16] 'TRANSFERINFO_CST' > *CPNAME
        < L[6] *CEPVAL
            < L[2]
                < A[6] 'LOT_ID' > * LOT_ID
                < A[6] 'lotID_01' > * LOT_ID value
            >
            < L[2]
                < A[13] 'CDAPURGE_FLAG' > * CDAPURGE_FLAG
                < A[3] ' YES ' > * CDAPURGE_FLAG value
            >
            < L[2]
                < A[11] 'BINDINGTYPE' > * BINDINGTYPE
                < U2 ' 0 ' > * BINDINGTYPE value
            >
            < L[2]
                < A[10] 'BINDINGKEY' > * BINDINGKEY
                < A[10] ' 1234567890 ' > * BINDINGKEY value
            >
            < L[2]
                < A[13] 'LIFE_FIX_FLAG' > * LIFE_FIX_FLAG
                < A[3] ' YES ' > * LIFE_FIX_FLAG value
            >
            < L[2]
                < A[11] 'CARRIERTYPE' > * CARRIERTYPE

```

< A[1] ' D ' > * CARRIERTYPE value

>

>

>

>

>

.

04.02.35 S2F50 Enhanced Remote Command Acknowledge**S2, F50 Enhanced Remote Command Acknowledge****S,H←E.**

Description: The equipment acknowledges the Enhanced Remote Command or reports any error(s). If the command is not accepted due to one or more invalid parameters (i.e. HCAACK=3), then a list of invalid parameters will be returned containing the parameter name and the reason for being invalid.

Structure:

```

L,2
  1.<HCAACK>
  2.L,n #of Parameter Groups
    1.L,2
      1.<CPNAME1>
      2.<CEPACK1>
    :
    n.L,2
      1.<CPNAMEn>
      2.<CEPACKn>

```

Exception:

If there are no invalid parameters, then a list of zero length will be sent for item 2.

TRANSFER Acknowledge:

```

'S2F50' /* Enhanced Remote Command Acknowledge E->H */
< L[2]
  < B[1] 4 > *HCAACK
  < L[0]>
>

```

04.02.36 S5F1 Alarm Report Send**S5, F1 Alarm Report Send****S,H←E, reply.**

Description: This message reports a change in or presence of an alarm condition. One message will be issued when the alarm is set and one message will be issued when the alarm is cleared. Irrecoverable errors and attention flags may not have a corresponding clear message.

Structure:

```

L, 3
  1.<ALCD>
  2.<ALID>
  3.<ALTX>

```

04.02.37 S5F2 Alarm Report Acknowledge**S5, F2 Alarm Report Acknowledge****S,H→E.**

Description: Acknowledge or error.

Structure:

<ACKC5>

04.02.38 S5F3 Enable/Disable Alarm**S5, F3 Enable/Disable Alarm****S,H→E, reply.**

Description: The purpose of this message is for the host to enable/disable alarm in reconciliation.

Structure:

L, 2

1.<ALED>

2.<ALID>

04.02.39 S5F4 Enable/Disable Alarm Acknowledge**S5, F4 Enable/Disable Alarm Acknowledge****S,H←E.**

Description: Acknowledge .

Structure:

<ACKC5>

04.02.40 S6F11 Event Report Send**S6, F11 Formatted Process Program Request****M,H←E, reply.**

Description: The purpose of this message is for the equipment to send group of reports to the host upon the occurrence of an event (CEID)

Whenever a particular event condition occurs, the Stocker Controller will send an S6F11 – Event Report Send message with the data associated with the event. Since many events may report the same data, common sets of data are defined and assigned a “Report ID (RPTID)”. The assignment of data or a RPTID to an event or CEID is known as “event report linking”. The SEMI standard provides the ability for the host (MCS) to programmatically define report contents and link reports to the various supported CEIDs. However, the Stocker Controller also has pre-defined reports and CEID linkage for the case where the

MCS does not choose to define its own reports.

Structure:

L,3

1.<DATAID>

2.<CEID>

3.L,a # "a" is always 1 for this system.

1. L,2

1.<RPTID>

2.L,b

1.<V1> #Variable data 1

:

b.<Vb> #Variable data b

04.02.41 S6F12 Event Report Acknowledge

S6, F12 Event Report Acknowledge

S,H→E.

Description: Acknowledge or error.

Structure:

<ACKC6>

04.02.42 S9F1 Unrecognized Device ID

S9, F1 Unrecognized Device ID

S,H←E.

Description: The device ID in the message block header did not correspond to any known device ID in the node detecting the error.

Structure:

<MHEAD>

04.02.43 S9F3 Unrecognized Stream Type

S9, F3 Unrecognized Stream Type

S,H←E.

Description: The equipment does not recognize the stream type in the message block header.

Structure:

<MHEAD>

04.02.44 S9F5 Unrecognized Function Type**S9, F5 Unrecognized Stream Type****S,H←E.**

Description: This message indicates that the function in the message ID is not recognized by the receiver.

Structure:

< MHEAD>

04.02.45 S9F7 Illegal Data**S9, F7 Illegal Data****S,H←E.**

Description: This message indicates that the stream and function were recognized, but the associated data format could not be interpreted.

Structure:

< MHEAD>

04.02.46 S9F9 Transaction Timer Timeout**S9, F9 Transaction Timer Timeout****S,H←E.**

Description: This message indicates that a transaction (receive) timer has timed out and that the corresponding transaction has been aborted. It is up to the host to respond to this error in an appropriate manner to keep the system operational.

Structure:

< SHEAD>

04.02.47 S9F11 Data Too Long**S9, F11 Data Too Long****S,H←E.**

Description: This message to the host indicates that the equipment has been sent more data than it can handle.

Structure:

< MHEAD>

04.02.48 S10F3 Terminal Display, Signal**S10, F3 Terminal Display, Single****S,H→E, [reply].**

Description: Data to be displayed.

Structure:

L, 2

1. < TID >
2. < TEXT>

04.02.49 S10F4 Terminal Display, Signal Acknowledge

S10, F4 Terminal Display, Single Acknowledge

S,H←E.

Description: Acknowledge or error.

Structure:

<ACKC10>

5. SECS Stream 功能情境(SECS Stream Function Scenario)

05.01 On Line Scenario (Off Line → On Line by Equipment)

Host	SxFx		SxFx	Equipment
		←	S1F13	Connect Request
Connect Request Acknowledge	S1F14	→		
		←	S6F11	Event Report Send(CEID=2 or 3) (Change On-Line Remote/Local Mode)
Event Report Acknowledge	S6F12	→		

05.02 Off Line Scenario (On Line → Off Line by Equipment)

Host	SxFx		SxFx	Equipment
		←	S6F11	Event Report Send(CEID=1) (Change Off-Line Mode)
Event Report Acknowledge	S6F12	→		

05.03 On Line Scenario (Off Line → On Line by Host)

Host	SxFx		SxFx	Equipment
Connect Request	S1F13	→		
		←	S1F14	Connect Request Acknowledge
Request On-line	S1F17	→		
		←	S1F18	ON-LINE Acknowledge
		←	S6F11	Event Report Send(CEID=2) (Change On-Line Remote Mode)
Event Report Acknowledge	S6F12	→		

05.04 On Line Scenario (On Line → Off Line by Host)

Host	SxFx		SxFx	Equipment
Offline Request	S1F15	→		
		←	S1F16	Offline Request Acknowledge
		←	S6F11	Event Report Send(CEID=1) (Change Off-Line Mode)
Event Report Acknowledge	S6F12	→		

05.05 On Line Scenario (On Line Local ↔ On Line Remote)

Host	SxFx		SxFx	Equipment
		←	S6F11	Event Report Send(CEID=2 or 3) (Change Online-Remote/Local Mode)
Event Report Acknowledge	S6F12	→		

05.06 Transport

Host	SxFx		SxFx	Equipment
Enhanced Remote Command	S2F49	→		
		←	S2F50	Enhanced Remote Command Acknowledge
		←	S6F11	Event Report Send(TransferInitiated(CEID=108))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(Transferring(CEID=111))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(TransferCompleted (CEID=107))
Event Report Acknowledge	S6F12	→		

05.07 Transport Cancel/Abort

Host	SxFx		SxFx	Equipment
Host Command Send (“CANCEL/ABORT”)	S2F41	→		
		←	S2F42	Host Command Acknowledge
		←	S6F11	Event Report Send(Initiated(CEID=106/103))
Event Report Acknowledge	S6F12	→		
CANCEL/ ABORT Completed				
		←	S6F11	Event Report Send(Completed (CEID=104/101))
Event Report Acknowledge	S6F12	→		
CANCEL/ ABORT Failed				
		←	S6F11	Event Report Send(Failed (CEID=105/102))
Event Report Acknowledge	S6F12	→		

05.08 Alarm Enable/Disable

Host	SxFx		SxFx	Equipment
Enable/disable Alarm	S5F3	→		
		←	S5F4	Enable/Disable Alarm Acknowledge

05.09 Alarm Report

Host	SxFx		SxFx	Equipment
		←	S5F1	Alarm Report Send
Alarm Report Acknowledge	S5F2	→		

05.10 Define/Link/Enable Reports

Host	SxFx		SxFx	Equipment
Disable All Events	S2F37	→		
		←	S2F38	
Delete All Reports	S2F33	→		
		←	S2F34	
Define Report	S2F33	→		
		←	S2F34	Define Report Acknowledge
Link Event Report	S2F35	→		

		←	S2F36	Link Event Report Acknowledge
Enable Event Report	S2F37	→		
		←	S2F38	Enable Event Report ACK

05.11 Install

Host	SxFx		SxFx	Equipment
When carrier install in controller				
		←	S6F11	Event Report Send(Install (CEID=151))
Event Report Acknowledge	S6F12	→		

05.12 Remove

Host	SxFx		SxFx	Equipment
When carrier Remove in controller				
		←	S6F11	Event Report Send(Install (CEID=152))
Event Report Acknowledge	S6F12	→		

05.13 ID Read

Host	SxFx		SxFx	Equipment
When carrier ID Read				
		←	S6F11	Event Report Send(ID Read (CEID=251))
Event Report Acknowledge	S6F12	→		
When carrier ID Read fail				
		←	S6F11	Event Report Send(ID Read (CEID=251))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(Remove old carrier (CEID=152))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(Install unknow ID carrier (CEID=151))
Event Report Acknowledge	S6F12	→		
When carrier ID Read mismatch				
		←	S6F11	Event Report Send(ID Read (CEID=251))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(Remove old carrier

				(CEID=152))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(Install new carrier (CEID=151))
Event Report Acknowledge	S6F12	→		

05.14 Transport with ID mismatch or ID Read Failed

Host	SxFx		SxFx	Equipment
Enhanced Remote Command	S2F49	→		
		←	S2F50	Enhanced Remote Command Acknowledge
		←	S6F11	Event Report Send(TransferInitiated(CEID=108))
Event Report Acknowledge	S6F12	→		
When carrier ID Read				
		←	S6F11	Event Report Send(ID Read (CEID=251))
Event Report Acknowledge	S6F12	→		
When carrier ID Read fail				
		←	S6F11	Event Report Send(TransferCompleted (CEID=107)) Carrier ID = unknown ID Result Code =5
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(Remove old carrier (CEID=152))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(Install unknown ID carrier (CEID=151))
Event Report Acknowledge	S6F12	→		
When carrier ID Read mismatch				
		←	S6F11	Event Report Send(TransferCompleted (CEID=107)) Carrier ID = new carrier Result Code =4
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send(Remove old carrier (CEID=152))
Event Report Acknowledge	S6F12	→		

		←	S6F11	Event Report Send(Install new carrier (CEID=151))
Event Report Acknowledge	S6F12	→		

05.15 Change to AGV Mode Flow(virtual case)

Host	SxFx		SxFx	Equipment
		←	S6F11	Event Report Send (MGV Port report PortOutOfService(CEID=701))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send (AGV Port report PortInService (CEID=702))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send (AGV Port report AGV Mode Change Report(CEID=705))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send (MGV Port report AGV Mode Change Report (CEID=705))
Event Report Acknowledge	S6F12	→		

05.16 Change to MGV Mode Flow(virtual case)

Host	SxFx		SxFx	Equipment
		←	S6F11	Event Report Send (MGV Port report PortOutOfService(CEID=701))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send (AGV Port report PortInService (CEID=702))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send (AGV Port report AGV Mode Change Report(CEID=706))
Event Report Acknowledge	S6F12	→		
		←	S6F11	Event Report Send (MGV Port report AGV Mode Change Report

				(CEID=706))
Event Report Acknowledge	S6F12	→		
Terminal Display, Signal	S10F3	→		
		←	S10F4	Terminal Display, Signal Acknowledge

6. Events

06.01 CEID List

Please be aware the list for RPTID/CEID is for reference information only and the controller is supposed to comply with SEMI E30 to support the Dynamic Event Report Configuration, means allow the host to dynamically define the equipment event reporting setup.

Control Events			
CEID	RPTID	VID	Contents
1	2		OffLineModeChange Report
		1	EqpName
2	2		OnLineLocal ModeChange Report
		1	EqpName
3	2		OnLineRemoteModeChange Report
		1	EqpName
51	200		AlarmCleared Report
		12	Command ID
		38	Error ID
		910	Unit Info
52	201		AlarmSet Report
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		2	AlarmID
		38	Error ID
		910	Unit Info
53			SCAutoCompleted Report

	2		
		1	EqpName
54			SCAutoInitiated Report
	2		
		1	EqpName
55			SCPauseCompleted Report
	2		
		1	EqpName
56			SCPaused Report
	2		
		1	EqpName
57			SCPauseInitiated Report
	2		
		1	EqpName
101			TransferAbortCompleted Report
	103		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
102			TransferAbortFailed Report
	103		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
103			TransferAbortInitiated Report
	103		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
104			TransferCancelCompleted Report
	103		
		12	Command ID

		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
105	TransferCancelFailed Report		
	103		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
106	TransferCancelInitiated Report		
	103		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
107	TransferCompleted Report		
	104		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		35	Result Code
		6	CarrierZoneName
108	TransferInitiated Report		
	109		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		35	Result Code
		6	CarrierZoneName
		15	Dest
109	TransferPaused		
	110		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
110	TransferResumed		

	110		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
111			Transferring
	109		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
		15	Dest
151			CarrierInstallCompleted Report
	4		
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
		36	Handoff Type
152			CarrierRemovedCompleted Report
	4		
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
	36	Handoff Type	
154			CarrierResumed Report
	108		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
		15	Dest
	31	Crane ID	
155			CarrierStored Report
	5		
		4	Carrier ID
		5	Carrier Loc

		6	CarrierZoneName
156			CarrierStoredAlt Report
	109		
		12	Command ID
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
		15	Dest
157			ShelfStatusChange Report
	205		
		906	Zone Data
158			CarrierWaitIn
	5		
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
159			CarrierWaitOut
	5		
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
160			CarrierTransferring
	5		
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
201			Crane/Vehicle Active
	11		
		12	Command ID
		31	Crane/Vehicle ID
		4	Carrier ID
202			Crane/Vehicle Idle
	11		
		12	Command ID
		31	Crane/Vehicle ID
		4	Carrier ID

205			Crane/Vehicle OutOfService
	207		
		31	Crane/Vehicle ID
206			Crane/Vehicle InService
	207		
		31	Crane/Vehicle ID
210			Crane/Vehicle AcquireStarted
	7		
		31	Crane/Vehicle ID
		4	CarrierID
		33	TransferPort
211			Crane/Vehicle AcquireCompleted
	7		
		31	Crane/Vehicle ID
		4	CarrierID
		33	TransferPort
212			Crane/Vehicle DepositStarted
	7		
		31	Crane/Vehicle ID
		4	CarrierID
		33	TransferPort
213			Crane/Vehicle DepositCompleted
	7		
		31	Crane/Vehicle ID
		4	CarrierID
		33	TransferPort
220			Crane/Vehicle Assigned
	8		
		31	Crane/Vehicle ID
		12	Command ID
221			Crane/Vehicle Unassigned
	8		
		31	Crane/Vehicle ID
		12	Command ID
222			VehicleDeparted
	9		

		31	Crane/Vehicle ID
		33	TransferPort
223			VehicleArrived
	9		
		31	Crane/Vehicle ID
		33	TransferPort
230			VehicleChargeStarted
	6		
		31	Crane/Vehicle ID
		32	ChargerID
231			VehicleChargeEnd
	6		
		31	Crane/Vehicle ID
		32	ChargerID
241			VehicleCoordinateChanged
	10		
		31	Crane/Vehicle ID
		29	VehicleCurrentPositionX
		30	VehicleCurrentPositionY
251			CarrierIDRead Report
	26		
		4	Carrier ID
		5	Carrier Loc
		34	IDReadState
252			ZoneCapacityChange
	112		
		906	ZoneData
701			PortOutOfService Report
	12		
		9	Port ID
702			PortInService Report
	12		
		9	Port ID
703			PortTypeInput Report
	12		

		9	Port ID
704			PortTypeOutput Report
	12		
		9	Port ID
705			AGV Mode Change Report
	12		
		9	Port ID
706			MGV Mode Change Report
	12		
		9	Port ID
602			EqLoadReq Report
	12		
		9	Port ID
603			EqUnloadReq Report
	12		
		9	Port ID
604			EqNoReq Report
	12		
		9	Port ID
800			CDAPurgeStart Report
	5		
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
801			CDAPurgeEnd Report
	5		
		4	Carrier ID
		5	Carrier Loc
		6	CarrierZoneName
810			OperatorInitiatedAction
	117		
		12	Command ID
		37	Command Type
		4	Carrier ID

		14	Source
		15	Dest
		13	Priority

06.02 SVID List (only Query for S1F3)

SVID	Variable ID Name	Format
2001	ControlState	<ControlState>
2002	SCState	<SCState>
2003	AlarmSet	L[n] <AlarmID>
2004	EnhancedCarriers	L[n] <EnhancedCarrierInfo>
2005	CurrentPortStates	L[n] <PortInfo>
2006	EnhancedTransfers	L[n] <EnhancedTransferCmd>
2007	CurrEqPortStatus	L[n] <EqPortInfo>
2008	EnhancedActiveZones	L[n] <ZoneData>
2009	PortTypes	L[n] <PortTypeInfo>
2010	ActiveVehicles	L[n] <VehicleInfo>
2011	VehiclePositions	L[n] <VehiclePositions>
2012	ReservedStorageInfos	L[n] <RESERVETOTALINFOS>
3001~5000	For Eqp define	

06.03 VID List

This is chart of variable data which includes all the previously defined types of variables

VID	Variable ID Name	Format	Data Description
1	EqpName	A	EqpName
2	AlarmID	A	Alarm ID

3	ControlState	U2	0 = Equipment OFF-LINE 1 = Attempt ON-LINE 2 = Host OFF-LINE 3 = ON-LINE/LOCAL 4 = ON-LINE/REMOTE
4	Carrier ID	A	Carrier ID
5	Carrier Loc	A	Carrier Location
6	CarrierZoneName	A	Carrier Zone Name
7	InstallTime	A16	Time <yyyyMMddHHmmssff>
8	CarrierState	U2	0: Wait in 1: Transferring 2: Completed 3: Alternate 4: Wait out 5: Installed
9	Port ID	A	<A[1..64]> Port ID
10	PortTransferState	U2	0=OutOfService, 1=InService
11	TransferState	U2	0=Queued, 1=Transferring, 2=Paused, 3=Canceling, 4=Aborting
12	CommandID	A	Command ID
13	Priority	U2	1~99
14	Source	A	Source ID
15	Dest	A	Dest ID
16	NextDest	A	NextDest ID
17	SCState	U2	0=Init, 1=Paused, 2=Auto, 3=Pausing
18	EqReqSatus	U2	0: no REQ 1: Load REQ on 2: Unload REQ on
19	EqPresenceStatus	U2	0: No presence 1: Presence

20	ZoneName	A	ZoneName
21	ZoneCapacity	U2	ZoneCapacity
22	ZoneType	U2	0: Shelf, 1: Port, 2: Other, 9: HandOff
23	ShelfId	A	ShelfId
24	DisabledStatus	U2	0: Enabled, 1: Disabled
25	PortTypeInfo	List	L[2] <PortID> <PortUnitType>
26	PortUnitType	U2	0: Input 1: Output
27	VehicleID	A	VehicleID
28	VehicleState	U2	0 = REMOVED 1 = NOT ASSIGNED 2 = ENROUTE 3 = PARKED 4 = ACQUIRING 5 = DEPOSITING
29	VehicleCurrentPositionX	A	VehicleCurrentPositionX
30	VehicleCurrentPositionY	A	VehicleCurrentPositionY
31	Crane/Vehicle ID	A	Crane/Vehicle ID
32	ChargerID	A	ChargerID
33	TransferPort	A	TransferPort
34	IDReadState	U2	Carrier ID Read State 0 : Success 1 : Failure 2 : Duplicate 3 : Mismatch 4 : No Carrier
35	Result Code	U2	0= Successful 1 = Other errors 2 = Zone is full 3 = Duplicate ID 4 = ID mismatch 5 = ID Read Failed 6 = Source Empty

			10 = Error But Carrier Already in Dest 64 = Interlock Error 100~ For Eqp define
36	Handoff Type	U2	0>manual, 1=automated
37	Command Type	U2	0=Transfer, 1=Cancel, 2=Abort
38	Error ID	A	ID of Error Ex : "DestOccupied" "SourceEmpty" "IDReadFail"
39	UnitID	A	UnitID
40	UnitState	U2	0=Normal 1=Double Storage 2=Empty Retrieval 3=Error Unit 4=Error Mode
41	Recovery Option	A	"ABORT"
42	ZoneTotalSize	U2	ZoneTotalSize
900	EnhancedCarrierInfo	List	<L[5] <CarrierID> <CarrierLoc> <CarrierZoneName> <InstallTime> <CarrierState> >
901	PortInfo	List	<L[2] <PortID> <PortTransferState>>

902	EnhancedTransferCmd	List	L[3] <TransferState> <CommandInfo> <TransferInfo>
903	CommandInfo	List	L[2] <CommandID> <Priority>
904	TransferInfo	List	L[4] <CarrierID> <Source> <Dest> <NextDest>
905	EqPortInfo	List	<L[4] <PortID> <PortTransferState> <EqReqSatus> <EqPresenceStatus> >
906	ZoneData	List	L[5] <ZoneName> <ZoneCapacity> <ZoneTotalSize> <ZoneType> <LocationStatus>
907	LocationStatus	List	L[n] < L[4] <ShelfId> <Carrier ID (Can be Empty)> <DisabledStatus > <ProcessState> >
908	VehicleInfo	List	L[3] <VehicleID> <VehicleState> <VehocleBattery>

909	VehiclePositions	List	L[3] <VehicleID > <VehicleCurrentPositionX> <VehicleCurrentPositionY>
910	UnitInfo	List	L[2] <UnitID> <UnitState>
911	WaitingTime	U2	ms 0 = Disable >0 = Enable
912	RESERVETOTALINFOS	List	L[n] [L2] <TargetLocationNmae> L[n] <CarrierID>
913	TargetLocationNmae	A	Port ID or Zone Name or LocationName
917	VehicleBattery	F4	Vehicle Battery(%)
914	TRANSFERINFO_CST	List	L[5] <LOT_ID> <CDAPURGE_FLAG> <BINDINGTYPE> <BINDINGKEY> <LIFE_FIX_FLAG> <CARRIERTYPE>
915	LOT_ID	A	Lot ID
916	CDAPURGE_FLAG	A	“YES” or “NO” YES means do the CDA PURGE NO means don’ t do the CDA PURGE
918	BINDINGTYPE	U2	0: Bind by EQP 1: Can’ t Bind 2: Bind

919	BINDINGKEY	A	When EQP get the same key , EQP will binding the same key Carrier
920	LIFE_FIX_FLAG	A	“YES” or “NO” YES means do the LIFE FIX NO means don’ t do the LIFE FIX
921	ProcessState	U2	0= None/Dynamic(無限制) 1 = Normal (DCS 可派送，MCS 異常處理不可派送) 2 = Abnormal (DCS 不可派送，MCS 異常處理可派送)
922	CARRIERTYPE	A	Enumeration by DCS
			UNKNOWN or ""
			B
			C
			D
			E
			F
			G
			I
			M
			R
			T

1001~2000 For Eqp define

06.04 RCMD List

RCMD	Parameter	Contents
CANCEL	L,2 1.<CANCEL> 2.L,1 1.L,2 1.<COMMANDID> 2.<\$COMMANDID>	Terminate all activity of a specific TRANSFEREXT command based on Command ID while the command is in the QUEUED state.
ABORT	L,2 1.<ABORT> 2.L,1 1.L,2 1.<COMMANDID> 2.<\$COMMANDID>	Abort the specific TRANSFEREXT command based on Command ID while the command is in Transferring state.
PAUSE	L,2 1.<PAUSE> 2.L,0	Put the SC into the PAUSING state.
RESUME	L,2 1.<RESUME> 2.L,0	Put the SC into the AUTO state.
SCAN	L,2 1.<SCAN> 2.L,1 1.L,2 1.<CARRIERLOC> 2.<\$CARRIERLOC>	To identify a carrier (If STK has on-board BCR/RFID).
PRIORITYUPDATE	L,2 1.<PRIORITYUPDATE> 2.L,2 1.L,2 1.<COMMANDID> 2.<\$COMMANDID> 2.L,2 1.<PRIORITY> 2.<\$PRIORITY>	Specify command ID to update the carrier move priority.

PORTTYPECHG	L,2 1.<PORTTYPECHG> 2.L,2 1.L,2 1.<PORTID> 2.<\$PORTID> 2.L,2 1.<PORTUNITTYPE> 2.<\$PORTUNITTYPE>	Specify port ID to change the direction
INSTALL	L,2 1.<INSTALL> 2.L,2 1.L,2 1.<CARRIERID> 2.<\$CARRIERID> 2.L,2 1.<CARRIERLOC> 2.<\$CARRIERLO >	Install carrier
REMOVE	L,2 1.<REMOVE> 2.L,2 1.L,2 1.<CARRIERID> 2.<\$CARRIERID> 2.L,2 1.<CARRIERLOC> 2.<\$CARRIERLOC>	Remove carrier
CDAPURGE	L,2 1.<CDAPURGE> 2.L,1 1.L,2 1.<CARRIERID> 2.<\$CARRIERID>	Purge process

RESERVE	L,2 1.<RESERVE> 2.L,3 1.L,2 1.<PRIORITY> 2.<\$PRIORITY> 2.L,2 1.<CARRIERLOC> 2.<\$CARRIERLOC> 3.L,2 1.<WaitingTime> 2.<\$WaitingTime>	Reserve AGV/LIFT/CRANE... to the location and waiting transfer command until WaitingTime has passed. EQP should compare the priority with other Transfer command ,then executing the high priority command. If the transfer commands priority is same as this command,EQP should executing Reserve command first.
RESERVESTORAGE	L,2 1.<RESERVESTORAGE> 2.L,2 1.L,2 1.<TargetLocationNmae> 2.<\$TargetLocationNmae> 2.L,2 1.<CarrierID> 2.<\$CarrierID>	When MCS Download transfer command to First EQP MCS will reserve target location (port or zone). When EQP receive this command , EQP should keep the target location clear until transfer carrier move in.
CANCELRESERVESTORAGE	L,2 1.<CANCELRESERVESTORAGE > 2.L,2 1.L,2 1.<TargetLocationNmae> 2.<\$TargetLocationNmae> 2.L,2 1.<CarrierID> 2.<\$CarrierID>	When MCS Cancel transfer command, MCS will cancel reserve target location (port or zone). When EQP receive this command , EQP should release the target location reserve state.

TRANSFER	TRANSFER /*Only use for Enhanced Remote Command*/ L4 1. DATAID 2. OBJSPEC 3.<TRANSFER> 4.L,3 1.L,2 1.<COMMANDINFO> 2.<\$ COMMANDINFO> 2.L,2 1.<TRANSFERINFO> 2.<\$TRANSFERINFO'> 3.L,2 1.<TRANSFERINFO_CST> 2.<\$TRANSFERINFO_CST'>	When MCS Download Transfer Command to EQP , EQP Should Follow CMD to transfer Carrier form SOURCE to DEST
NOTTRANSFERNOTIFY	L,2 1.< NOTTRANSFERNOTIFY> 2.L,2 1.L,2 1.<CARRIERID> 2.<\$CARRIERID> 2.L,2 1.<CARRIERLOC> 2.<\$CARRIERLOC>	When MCS without Transfer Command to EQP EQP Should show information on touch penal

06.05 Message Data Dictionary

ID	Format	Size	Description
MDLN	A	Max:30	Equipment Model Type. Same data as returned by S1F2.
SOFTREV	A	Max:30	Software Revision.
SVID	U2		Status Variable ID.
SV			Status Variable value. The value associated with a specified SVID. See "SVID" below. Each SVID used in system is specified and the structure of the associated SV is explained.
SVNAME	A	Max:60	Status Variable Name
UNITS	A		Units Identifier refer by SEMI E5 Section 12.
COMMACK	B	1	Establish communications acknowledge code, 1 byte 0 = Accepted 1 = Denied, Try Again 2-63 Reserved
OFLACK	B	1	Acknowledge code for OFF-LINE request 0 = OFF-LINE acknowledge 1-63 Reserved
ONLACK	B	1	Acknowledge code for ON-LINE request 0 = ON-LINE accepted 1 = ON-LINE not allowed 2 = Equipment already ON-LINE 3-63 Reserved
ECID	U2		Equipment constant ID
ECV	A		Equipment Constant Value.
DATETIME	A	14	YYYYMMDDHHMMSS
ABS	B		Any binary string
ECNAME	A		Equipment Constant Name
ECMIN	U2		Equipment Constant Minimum Value
ECMAX	U2		Equipment Constant Maximum Value
ECDEF	U2		Equipment Constant Default Value
TIACK	B	1	Time Acknowledgement Code, 1 byte 0 = OK 1 = Error, Not done. 65 = System Error

DATAID	U4		Data ID
RPTID	U2		Report ID
VID	U2		Variable ID
DRACK	B	1	Defined Report Acknowledge Code, 1 byte 0 = Accepted 1 = Denied, Insufficient space. 2 = Denied, Invalid Format. 3 = Denied, At least one of RPTID already defined. 4 = Denied, At least one of VID does not exist 65 = System Error 67 = Denied, The same VID is defined in the scenario 68 = Denied, Specified RPTID does not exist
CEID	U2		Collected event ID
LRACK	B	1	Link Report Acknowledge Code, 1 byte 0 = Accepted 1 = Denied, Insufficient space 2 = Denied, Invalid Format 3 = Denied, At least one of CEID link Already defined 4 = Denied, At least one CEID does not exist 5 = Denied, At least one RPT ID does not exist 65 = System Error 67 = Denied, Same RPTID Already defined
CEED	Boolean		Collection Events Enable / Disable Code, 1 byte False = Disable True = Enable
ERACK	B	1	Enable / Disable Event Report Acknowledge Code, 1 byte 0 = Accepted 1 = Denied, At least one CEID does not exist. 65 = System Error
RCMD	A		Remote command code or string
CPNAME	A		Command Parameter Name
CPVAL	A		Command Parameter Value

HACK	B	1	Host Command Parameter Acknowledge Code, 1 byte 0 = Acknowledge, Command has been performed 1 = Command does not exist 2 = Cannot perform now. 3 = At least one parameter is invalid. 4 = Acknowledge, command will be perform with completion signaled later by an event 5 = Rejected, Already in desired condition 6 = Not such object exists 64 = CPNAME and CPVAL is insufficient 65 = System Error 70~FF For Eqp define
OBJSPEC	A		Object Specifier. Always the empty string for this system.
ALCD	B	1	Alarm Code bit8=1 : Alarm set bit8=0 : Alarm cleared bit7-1 : Alarm classification code 0= Not used
ALID	U2		Alarm ID
ALTX	A		Alarm Text
ALED	B	1	Alarm Enable / Disable Code, 1 byte bit8 = 1: Enable alarm bit8 = 0: Disable alarm
ACKC5	B	1	Acknowledge Code, 1byte 0 = Accepted 1 = Not accepted 65 = System Error
ACKC6	B	1	Acknowledge Code, 1byte 0 = Accepted 65 = System Error
MHEAD	B	1	Message Header In Error
SHEAD	B	1	Stored header related to the transaction timer.
TID	B	1	Terminal No., 1 byte 0 = Main Terminal
TEXT	B	1	A single line of characters

ACKC10	B	1	Acknowledge Code, 1byte 0 = Accepted for display 1 = Message will not be displayed 2 = Terminal not available 65 = System Error
CEPACK	U1 / List	-	Command Enhanced Parameter Acknowledge. If a specific value of CPNAME is defined to have a CEPVAL that is a LIST, then CEPACK shall have the same structure as the corresponding list format of CEPVAL as used in S2,F49. Otherwise CEPACK will be a 1 byte integer. Enumerated: 0 = No error 1 = Parameter name (CPNAME) does not exist 2 = Illegal value specified for CEPVAL 3 = Illegal format specified for CEPVAL 4 = Parameter name (CPNAME) not valid as used 5-63 Reserved

7. Other Rule 其他規則

07.01 Abnormal Cassette ID Naming rule

當 VCR 讀取失敗沒有 ID，Controller 上報 搬運失敗時 Carrier ID 的編碼邏輯，所有搬運設備有 VCR

都適用此規則，範例如下：

- Carrier ID : TEST1234
- 上報搬運失敗上報的 Carrier ID : UN00100001
 - UN 表示未知
 - 001 表示設備流水號(附錄 1)
 - 00001 表示為流水碼(00001~99999)

8. 附錄

1.設備流水號(實際編碼屆時以該廠域議定)

[illegible]